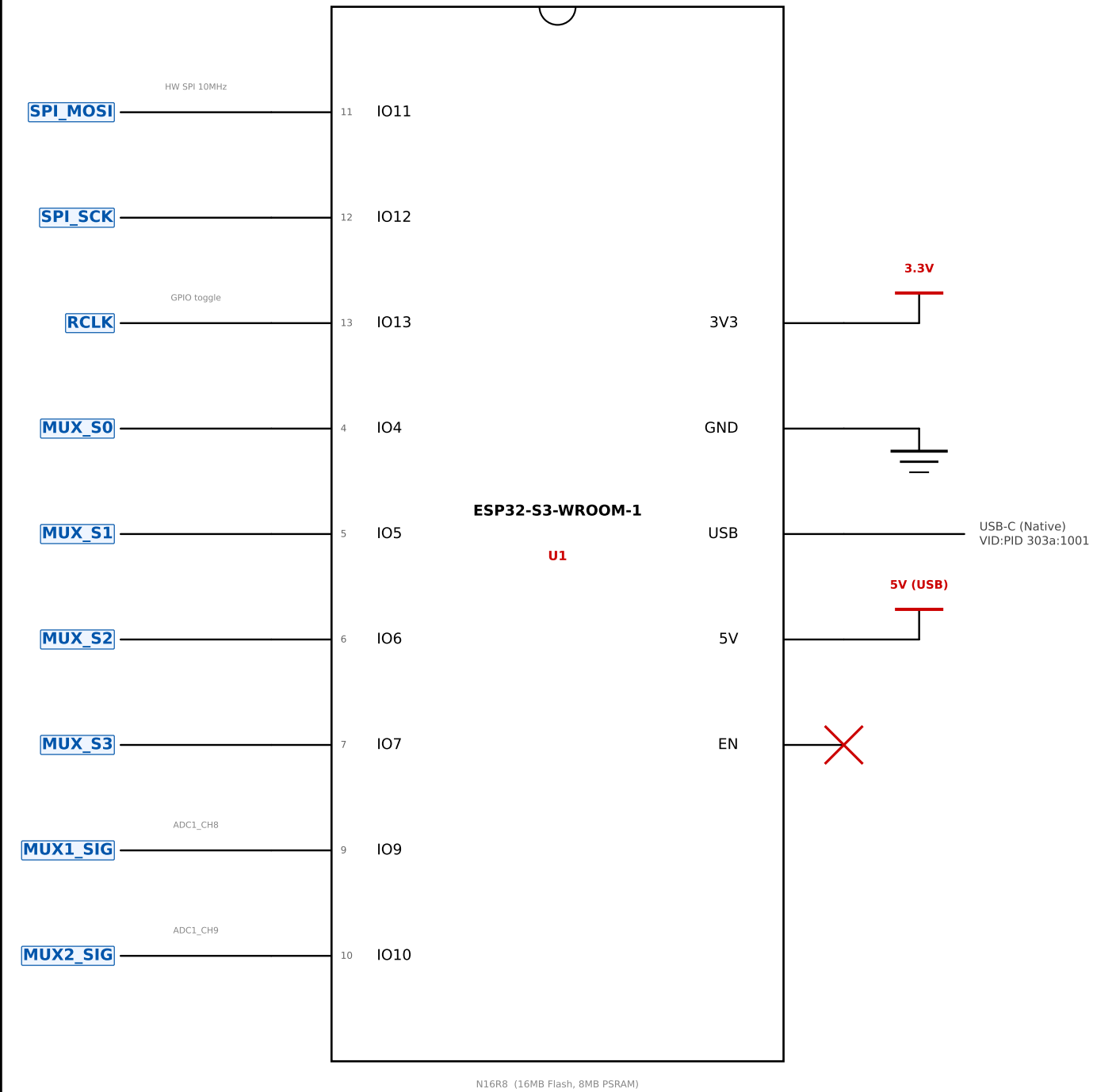


SHEET 1 — MCU, POWER DISTRIBUTION, BUS CONNECTIONS



GPIO PIN ASSIGNMENT & BUS ROUTING

GPIO	Net Name	Function	Destination
I011	SPI_MOSI	SPI Master Out	U2 pin 14 (SER) — cascades to U3-U6
I012	SPI_SCK	SPI Clock	U2-U6 pin 11 (SRCLK) — parallel
I013	RCLK	Latch (GPIO)	U2-U6 pin 12 (RCLK) — parallel
I04	MUX_S0	MUX Select bit 0	U7+U8 S0 — parallel
I05	MUX_S1	MUX Select bit 1	U7+U8 S1 — parallel
I06	MUX_S2	MUX Select bit 2	U7+U8 S2 — parallel
I07	MUX_S3	MUX Select bit 3	U7+U8 S3 — parallel
I09	MUX1_SIG	ADC1_CH8 input	U7 SIG (Columns 0-14)
I010	MUX2_SIG	ADC1_CH9 input	U8 SIG (Columns 15-29)

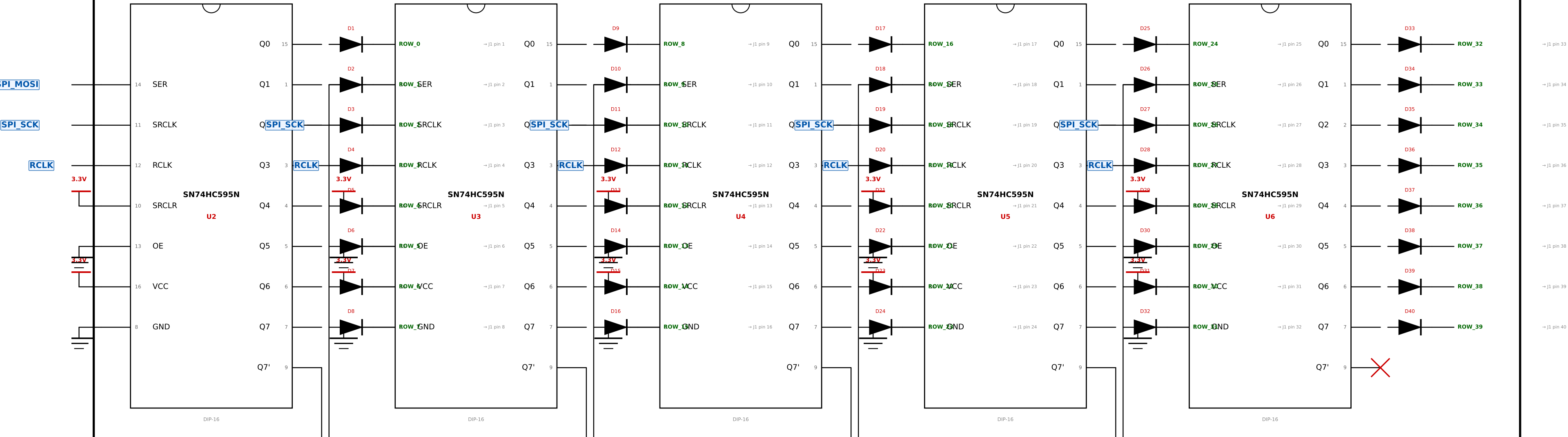
POWER DISTRIBUTION (3.3V RAIL)



40x30 PRESSURE SENSOR MATRIX

MCU + Power + Bus	
Date: 2026-04-08 Rev: 1.0	Sheet: 1 of 3 ESP32-S3 Breadboard Prototype
Components: U1: ESP32-S3 U2-U6: SN74HC595N U7-U8: CD74HC4067 D1-D40: 1N4148TA	R1-R2: 10K J1: FFC-40 J2: FFC-30

SHEET 2 — 74HC595 SHIFT REGISTER CASCADE + DIODE ARRAY (ACTIVE-HIGH ROW DRIVE)



NOTES — 595 CASCADE OPERATION

- SPI sends MSB first. Buffer layout: buf[0] = U6 (farthest, rows 32-39), buf[4] = U2 (closest, rows 0-7)
- Row select formula: chip = row / 8, bit = row % 8, buf[NUM_595 - 1 - chip] = (1 << bit)
- Only ONE row driven HIGH at a time. All other 595 outputs = LOW.
- Diodes D1-D40 (1N4148TA): Anode (no band) faces 595 output, Cathode (band) faces FFC row pin.
Purpose: prevent ghosting / sneak current paths through the resistive velostat matrix.
- After SPI transfer of 5 bytes: toggle RCLK (IO13) HIGH then LOW to latch all outputs simultaneously.
- 595 pin 10 (SRCLR) = 3.3V (clear disabled), pin 13 (OE) = GND (outputs always enabled).

40x30 PRESSURE SENSOR MATRIX

595 Cascade + Diode Array

Date: 2026-04-08

Rev: 1.0

Sheet: 2 of 3

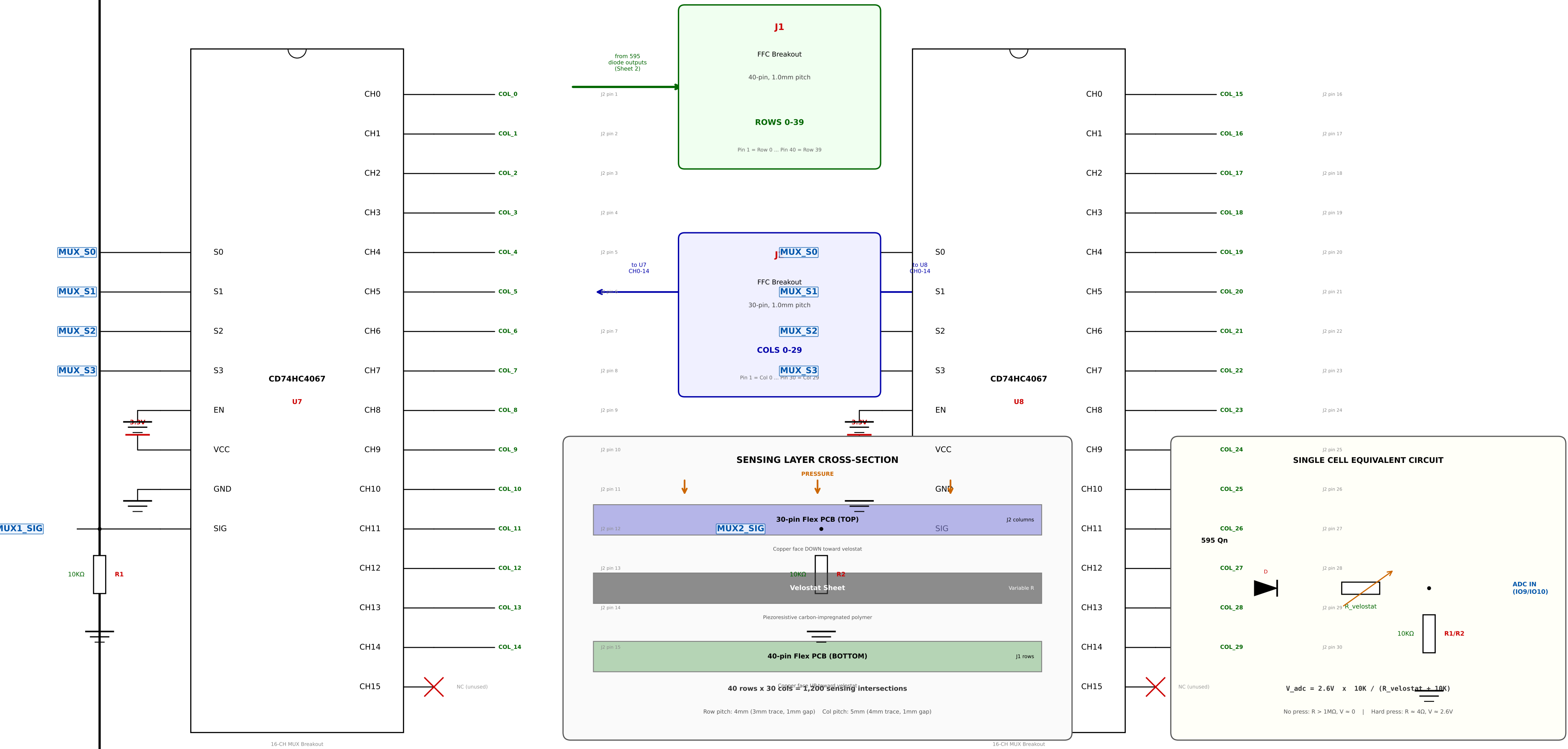
ESP32-S3 Breadboard Prototype

Components:

U1: ESP32-S3 U2-U6: SN74HC595N
U7-U8: CD74HC4067 D1-D40: 1N4148TA

R1-R2: 10K J1: FFC-40 J2: FFC-30

SHEET 3 — COLUMN MUX READERS + FFC CONNECTORS + SENSING LAYER



40x30 PRESSURE SENSOR MATRIX

MUX + Connectors + Sensing

Date: 2026-04-08
Rev: 1.0

Sheet: 3 of 3
ESP32-S3 Breadboard Prototype

Components:

U1: ESP32-S3 U2-U6: SN74HC595N
U7-U8: CD74HC4067 D1-D40: 1N4148TA

R1-R2: 10K J1: FFC-40 J2: FFC-30